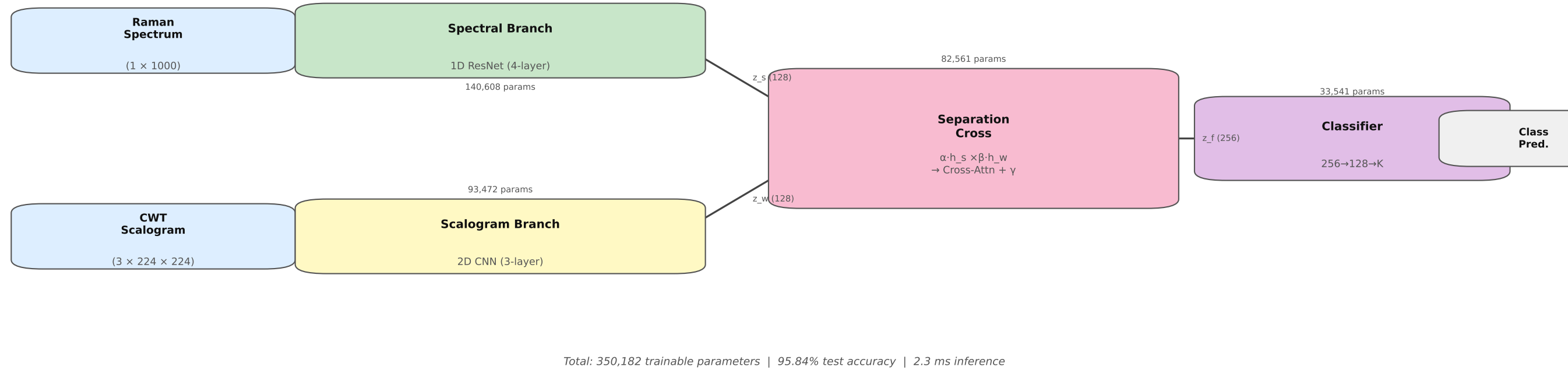
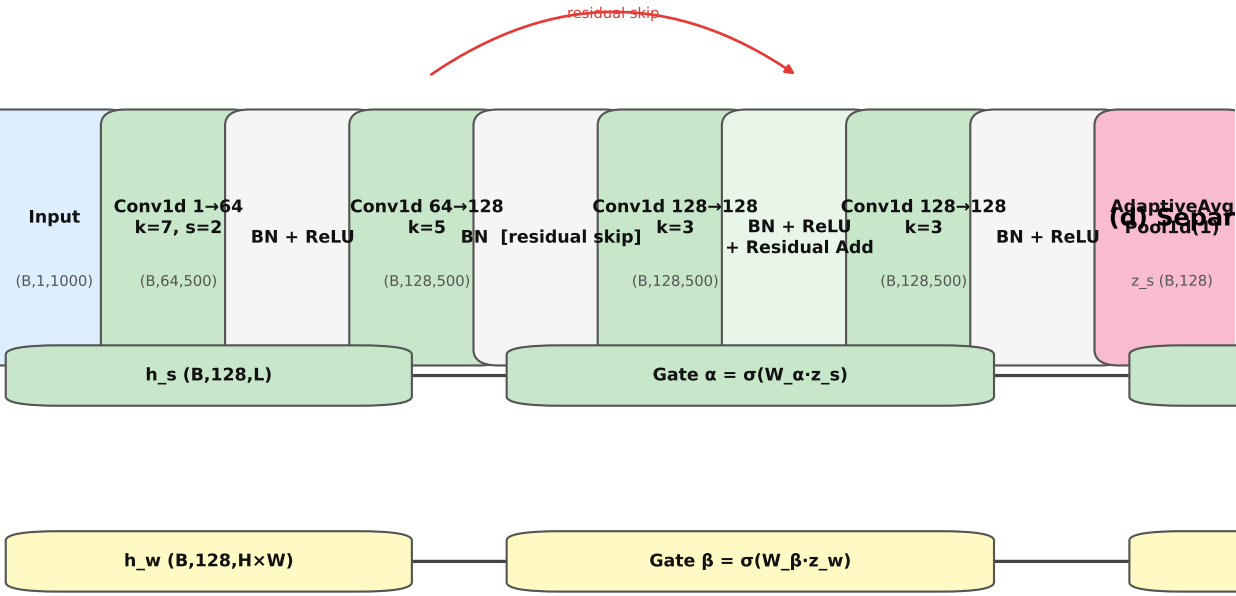


# PINNACLE Architecture: Dual-Stream Spectral-Scalogram Fusion

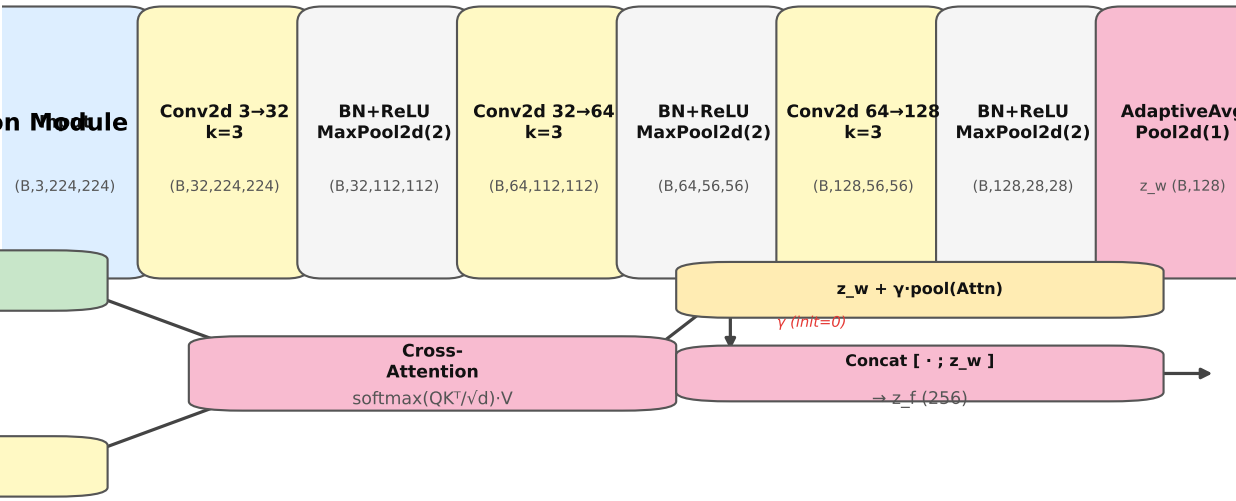
(a) PINNACLE: Dual-Stream Architecture



(b) SpectralBranch (1D ResNet)



(c) ScalogramBranch (2D CNN)



(d) SeparationCross Fusion Module

